



# Biodiversity risk and value creation in emerging markets

Wenjie Sun<sup>a</sup>, Jianhua Mei<sup>a</sup>, Kecun Chen<sup>a</sup>, Zhaoqi Li<sup>a,b,\*</sup>

<sup>a</sup> Seokyeong University, Seongbuk-gu, Seoul, 02713 South Korea

<sup>b</sup> College of Business, Gachon University, Seongnam 13120, South Korea

## ARTICLE INFO

### JEL Classification:

G32  
Q57  
O35  
G14

### Keywords:

Biodiversity risk management  
Green innovation  
Corporate value

## ABSTRACT

This study investigates how biodiversity risk management affects corporate value in Chinese A-share listed firms. Findings indicate that (1) proactive biodiversity risk management significantly enhances firm value, particularly in large enterprises and environmentally sensitive industries. (2) Green innovation is a crucial mechanism through which biodiversity risk management translates into corporate value enhancement. (3) Institutional investor oversight amplifies this positive effect, providing practical guidance for market-based biodiversity conservation strategies in emerging economies.

## 1. Introduction

Biodiversity loss (Chen et al., 2025; Ma et al., 2024; Wang, 2025), climate change (Adhikari et al., 2025; Cao et al., 2025; Lei and Xu, 2024), and energy resource depletion are triple environmental crises threatening global economic systems and corporate sustainability (Lei et al., 2025). The correlations between these challenges establish complex risk landscapes for firms as energy transition imperatives increasingly impact biodiversity conservation efforts. While climate and energy risk research has developed robust theoretical frameworks (Kuziboev et al., 2025; Marrero et al., 2015; Ongsakul et al., 2025; Yang and Song, 2025), biodiversity risk research remains underdeveloped, with a particular deficit in micro-level empirical evidence. Recent work addresses this gap. For example, using 25,979 Chinese firm-year observations, Zhou et al. (2025) demonstrate that biodiversity risk exposure reduces dividend payouts via financing constraints and resource reallocation effects. This particularly applies to firms with weak political ties and governance deficiencies as governance quality significantly affects environmental performance (Guo et al., 2024; Lin et al., 2024; Zhang and Han, 2025). Concurrently, Zhao et al. (2025) reveal that managerial myopia impedes biodiversity performance, demonstrating that reducing short-termism by one unit improves biodiversity outcomes by 0.83 %, with regulatory environments and innovation capacity acting as critical moderators. The Kunming-Montreal Global Biodiversity Framework has elevated the urgency of integrating biodiversity considerations into corporate risk management (Chen et al., 2025; He et al., 2025), and associated studies collectively highlight how biodiversity practices directly shape firms' financial decision-making and long-term value creation.

This study extends the traditional Porter hypothesis by examining market-driven risk perception mechanisms (Ambec et al., 2013). Drawing on dynamic capabilities' theory (Teece, 2018), we argue that biodiversity risk management stimulates the development of specific organizational capabilities that catalyze green innovation. This aligns with recent findings that uncertainty drives strategic resource reallocation that affect firms' growth and value (Bansal et al., 2019; Liu et al., 2025; Zhang et al., 2025). Additionally,

\* Corresponding author at: Seokyeong University, Seongbuk-gu, Seoul, 02713, South Korea.

E-mail addresses: [sunwenjie@skuniv.ac.kr](mailto:sunwenjie@skuniv.ac.kr) (W. Sun), [meijianhua@skuniv.ac.kr](mailto:meijianhua@skuniv.ac.kr) (J. Mei), [chenkecun@skuniv.ac.kr](mailto:chenkecun@skuniv.ac.kr) (K. Chen), [lizhaoqi@gachon.ac.kr](mailto:lizhaoqi@gachon.ac.kr) (Z. Li).

referring to attention-based theory (Ocasio, 1997), we propose that institutional investor attention amplifies corporate strategic responses to biodiversity risks, serving as a critical external governance mechanism.

This study systematically examines the impact pathway of corporate biodiversity risk identification and management capabilities on Chinese listed companies' value by constructing a natural language processing (NLP)-based biodiversity risk management metric. Referencing Giglio et al. (2023) and Kalhor and Ahmed (2025), we combine text analysis with the Bidirectional Encoder Representations from Transformers (BERT) model to develop a biodiversity risk management index. To address potential endogeneity, we employ instrumental variables (IVs), system generalized method of moments (GMM) estimation, and propensity score matching (PSM), among other econometric methods for robustness testing.

This study makes three core contributions to corporate finance and financial economics research. First, it breaks the traditional Porter hypothesis-centered analytical paradigm of government regulation. Based on dynamic capability theory and attention-based theory, we construct a market-driven biodiversity risk governance framework, revealing how firms' biodiversity risk identification enhances their organizational innovation capabilities and creates shareholder value. Second, the study defines institutional investor shareholding as a quasi-regulatory pressure source distinct from traditional government intervention. The findings reveal that institutional investor attention amplifies the value-creation effects of biodiversity risk management, in contrast to rigid regulatory compliance and representing a market-based governance mechanism similar to external stakeholder monitoring implying disciplinary pressure in corporate finance contexts (Lin and Guan, 2024). This establishes a market-intermediated corporate governance paradigm for environmental risk management. Third, we develop a biodiversity risk management index using the BERT model, addressing the measurement challenges in environmental finance research and greenwashing detection and offering firm-level evidence supporting risk-based valuation under the Kunming-Montreal Framework. These findings extend the theory of voluntary corporate disclosure in finance, demonstrating that in regulatory voids, institutional investor attention can activate firms' strategic risk awareness and value creation through informal governance mechanisms. This has significant implications for capital market development and corporate risk management in emerging economies seeking to integrate biodiversity considerations into financial decision-making.

## 2. Methodology

### 2.1. Variable construction

We use corporate value (TQ) as the dependent variable, which is calculated as the ratio of a firm's market value to the total book value of its assets. In robustness tests, we use the market-to-book ratio (MBR) as an alternative measure of corporate value.

The independent variable is the biodiversity risk management index (BD), constructed based on annual report text analysis. Following the established methodologies (Giglio et al., 2023; Kalhor and Ahmed, 2025), we first establish a biodiversity terminology dictionary, including keywords such as "biodiversity," "ecosystem," "species," "habitat," "forest," "wetland," "wildlife," and "coral." Second, we screen paragraphs from firms' annual reports containing at least two biodiversity-related sentences. Third, we apply the BERT model to conduct sentiment analysis on each selected sentence. Unlike traditional risk indicators, our metric reflects the degree to which companies identify and actively respond to biodiversity risks, rather than the level of risk exposure itself. Specifically, positive sentences (e.g., "the company actively addresses biodiversity challenges," "implements ecological protection measures") are assigned a value of +1, indicating higher risk management awareness, negative sentences (e.g., "ignores biodiversity impacts," "has not taken protective measures") are assigned a value of -1, indicating a lower risk management awareness, and neutral sentences are assigned a value of 0. Finally, we sum and standardize the sentiment values of all relevant sentences in each annual report to form the company's annual biodiversity risk management index.

### 2.2. Sample and model

The research sample includes Chinese A-share listed companies spanning 2013–2023, excluding financial and insurance industries and ST listed companies, yielding 10,876 firm-year observations. Data sources include the China Stock Market & Accounting Research database, the National Intellectual Property Administration patent database, and the National Forestry and Grassland Administration nature reserve data. The baseline regression model is as follows:

$$TQ_{i,t} = \alpha + \beta_1 BD_{i,t} + \gamma X_{i,t} + \mu_i + \lambda_t + \varepsilon_{i,t} \quad (1)$$

**Table 1**  
Variable definitions and measurement.

| Variables | Measurement                                                                                                                     |
|-----------|---------------------------------------------------------------------------------------------------------------------------------|
| TQ        | Tobin's Q, calculated as the ratio of a firm's market value to its total assets' book value                                     |
| BD        | Biodiversity risk management index, constructed based on textual analysis of annual reports using the BERT model                |
| Size      | Natural logarithm of total assets                                                                                               |
| Lev       | Financial leverage, calculated as total liabilities divided by total assets                                                     |
| Growth    | Revenue growth rate, calculated as the annual percentage change in operating revenue                                            |
| Age       | Natural logarithm of years since listing                                                                                        |
| Board     | Natural logarithm of the number of directors on the board                                                                       |
| INDEP     | Proportion of independent directors, calculated as the number of independent directors divided by the total number of directors |

where  $X_{i,t}$  represents the set of control variables (listed in Table 1) and  $\mu_i$  and  $\lambda_t$  represent company and year fixed effects (FEs), respectively.

### 3. Results

#### 3.1. Baseline results

Table 2 presents the baseline results. Column (1) includes only the BD variable, while column (2) introduces the control variables. The results without considering control variables reveal that BD is positively correlated with TQ. After adding control variables, the positive correlation between BD and TQ remains (coefficient of 0.147). This provides preliminary evidence for the link between active biodiversity risk management and corporate value, indicating that the market may positively evaluate corporate actions to identify, disclose, and actively address biodiversity risks.

#### 3.2. Robustness tests

We employ multiple methods to verify the robustness of our baseline results. Table 3 presents the results of three alternative estimation methods. Column (1) uses the MBR as an alternative measure of corporate value to reduce the impact of asset valuation biases on the results. Column (2) employs the system GMM method to address potential dynamic endogeneity issues, which effectively mitigates endogeneity concerns using lagged variables as IVs. The Hansen test results fail to reject the null hypothesis of IV validity, and the AR(2) test results show no serial correlation. Column (3) uses PSM to control for sample selection bias. To do so, we divide the sample into high and low risk groups according to the median biodiversity risk management for matching to eliminate potential selection bias. All robustness tests verify the positive association between biodiversity risk management and corporate value that are consistent with the baseline model, increasing the credibility of our research conclusions.

To further address endogeneity, we employ the IV method (Table 4), using the logarithm of the geographical distance between a company and the nearest national key ecological functional zone (DIST) as the IV (Li et al., 2024; Tan et al., 2025). This IV was chosen for two reasons. First, ecological functional zones have higher biodiversity levels, and companies geographically closer to these areas are more likely to face higher biodiversity-related regulatory and social pressure, prompting them to enhance their biodiversity risk identification and management capabilities (relevance condition). Second, ecological functional zone designation is primarily based on natural geographical and administrative division considerations, which is relatively independent of individual corporate value-creation processes. After controlling for regional FEs, the distance variable satisfies the exclusion requirement. The first-stage F-statistic is 42.56, which is greater than the Stock–Yogo weak IV critical value of 10, indicating that the IV satisfies the relevance requirement. The second-stage estimation results reveal a positive association between BD and TQ, further validating our baseline findings.

#### 3.3. Cross-sectional evidence

We next conduct group analyses based on firm size and industry characteristics to explore the heterogeneity in the relationship

**Table 2**  
Baseline regression.

| Variables      | (1)<br>TQ           | (2)<br>TQ            |
|----------------|---------------------|----------------------|
| BD             | 0.179***<br>(2.65)  | 0.147**<br>(2.18)    |
| Size           |                     | −0.428***<br>(−9.65) |
| Lev            |                     | −1.816***<br>(−8.32) |
| Growth         |                     | 0.213***<br>(4.83)   |
| Age            |                     | −0.082<br>(−1.46)    |
| Board          |                     | 0.128<br>(1.31)      |
| INDEP          |                     | 1.235**<br>(2.51)    |
| Constant       | 2.348***<br>(28.73) | 11.193***<br>(11.29) |
| Firm FE        | Yes                 | Yes                  |
| Year FE        | Yes                 | Yes                  |
| N              | 10,876              | 10,876               |
| R <sup>2</sup> | 0.126               | 0.375                |

Note: t-statistics are in parentheses, \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.1$ .

**Table 3**  
Robustness tests.

| Variable       | (1)<br>MBR        | (2)<br>TQ          | (3)<br>TQ         |
|----------------|-------------------|--------------------|-------------------|
| BD             | 0.225**<br>(2.31) | 0.156**<br>(2.06)  | 0.138**<br>(1.98) |
| L.TQ           |                   | 0.175***<br>(4.53) |                   |
| Controls       | Yes               | Yes                | Yes               |
| Firm           | Yes               | Yes                | Yes               |
| Year           | Yes               | Yes                | Yes               |
| N              | 10,876            | 9788               | 9262              |
| R <sup>2</sup> | 0.358             | 0.482              | 0.363             |
| Hansen p-value |                   | 0.213              |                   |
| AR(2) p-value  |                   | 0.328              |                   |

Note: t-statistics are in parentheses, \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.1$ .

**Table 4**  
Instrumental variable regression.

| Variables      | (1)<br>BD           | (2)<br>TQ          |
|----------------|---------------------|--------------------|
| DIST           | -0.348***<br>(6.52) |                    |
| IV_BD          |                     | 0.183***<br>(3.97) |
| Controls       | Yes                 | Yes                |
| City           | Yes                 | Yes                |
| Firm FE        | Yes                 | Yes                |
| Year FE        | Yes                 | Yes                |
| N              | 10,876              | 10,876             |
| F-statistic    | 42.56               |                    |
| R <sup>2</sup> | 0.302               | 0.376              |

Note: t-statistics are in parentheses, \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.1$ .

between biodiversity risk management and corporate value (Table 5). We divide the sample into large and small- and medium-sized enterprises based on the median of total assets. Results show that the impact of BD on TQ is relatively more pronounced for large enterprises, with a chi-square test result of 4.12 ( $p < 0.10$ ) for the difference between groups. This may be attributable to large enterprises possessing more abundant resources to address biodiversity risks, enabling them to transform risk identification into strategic opportunities more effectively.

For the industry dimension, we refer to the Industry Classification Guidelines for Listed Companies revised by the China Securities Regulatory Commission in 2012, the Listed Company Environmental Protection Inspection Industry Classification Management Directory produced by the Ministry of Environmental Protection in 2008, and the Listed Company Environmental Information Disclosure Guidelines. The classification includes 16 industries covering coal, mining, textiles, leather, paper, petrochemicals, pharmaceuticals, chemicals, metallurgy, and thermal power as high environmental sensitivity industries. The results indicate that the impact of BD on TQ appears to be more significant in high environmental sensitivity industries, with a chi-square test result of 5.86 ( $p < 0.10$ ) for the difference between groups, indicating that the market assigns a higher premium to biodiversity risk management in environmentally sensitive industries, consistent with the risk premium differentiation theory. The heterogeneity analysis results emphasize the important influence of company characteristics and industry context on the relationship between biodiversity risk management and corporate value, providing a theoretical basis for companies to develop targeted risk management strategies.

**Table 5**  
Heterogeneity analysis.

| Variables      | Large Firms<br>TQ | Small-Medium Firms<br>TQ | High Environmental Sensitivity<br>TQ | Low Environmental Sensitivity<br>TQ |
|----------------|-------------------|--------------------------|--------------------------------------|-------------------------------------|
| BD             | 0.205**<br>(2.38) | 0.109*<br>(1.76)         | 0.226**<br>(2.42)                    | 0.103*<br>(1.85)                    |
| Controls       | Yes               | Yes                      | Yes                                  | Yes                                 |
| Firm FE        | Yes               | Yes                      | Yes                                  | Yes                                 |
| Year FE        | Yes               | Yes                      | Yes                                  | Yes                                 |
| N              | 3916              | 6960                     | 3223                                 | 7653                                |
| R <sup>2</sup> | 0.405             | 0.347                    | 0.423                                | 0.359                               |

Note: t-statistics are in parentheses, \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.1$ .

#### 4. Pathways and interactions

To explore the intrinsic mechanisms through which corporate biodiversity risk management capabilities affect corporate value, we test the transmission pathway of green innovation (GI) and the moderating effect of institutional investor attention (IA). GI is measured by the natural logarithm of the number of enterprises' green patent applications (Lei and Xu, 2025), and IA is represented by the proportion of institutional investor shareholdings. We construct the following models:

$$GI_{i,t} = \alpha_1 + \beta_2 BD_{i,t} + \gamma_1 X_{i,t} + \mu_{1i} + \lambda_{1t} + \varepsilon_{1i,t} \quad (2)$$

$$TQ_{i,t} = \alpha_2 + \beta_3 BD_{i,t} + \beta_4 GI_{i,t} + \gamma_2 X_{i,t} + \mu_{2i} + \lambda_{2t} + \varepsilon_{2i,t} \quad (3)$$

$$GI_{i,t} = \alpha_3 + \beta_5 BD_{i,t} + \beta_6 IA_{i,t} + \beta_7 BD_{i,t} \times IA_{i,t} + \gamma_3 X_{i,t} + \mu_{3i} + \lambda_{3t} + \varepsilon_{3i,t} \quad (4)$$

The results in Table 6 indicate that BD has a positive impact on GI (coefficient 0.085), and GI has a certain positive impact on TQ after controlling for BD (coefficient 0.418). These results support GI as a transmission channel connecting biodiversity risk management with corporate value. Column (3) reveals that the coefficient of the  $BD \times IA$  interaction term is positive and significant at the 10 % level, indicating that institutional investor attention may influence the relationship between BD and GI. Overall, these findings support the possible existence of a risk management capability–innovation response–value creation transmission chain.

#### 5. Conclusion

This study contributes to environmental economics by examining how biodiversity risk management affects firm value in Chinese capital markets. Our findings reveal that proactive biodiversity risk management significantly enhances corporate value. This relationship exhibits significant heterogeneity, with stronger effects for large enterprises and environmentally sensitive industries, indicating contingent valuation effects aligned with risk exposure differentials. GI functions as a critical mechanism through which risk management improves value, and institutional IA amplifies this relationship through external governance pressure. These findings extend beyond the traditional Porter hypothesis by demonstrating that market-driven environmental risk perception can stimulate value-creating innovation absent stringent regulatory mandates. Furthermore, our methodology advances the measurement of corporate biodiversity risk management using sophisticated NLP techniques, addressing the limitations of conventional environmental metrics that often fail to capture substantive management practices beyond symbolic disclosure.

Our results offer several actionable insights for practitioners and policymakers. First, biodiversity risk management is not merely a compliance burden but also a potential source of competitive advantage in an increasingly eco-conscious market. The positive effects demonstrate that capital markets reward firms that effectively manage biodiversity risks, establishing financial incentives that align profit motives with conservation objectives. Second, our findings highlight the importance of institutional investor engagement as a market-based governance mechanism, indicating that investor-focused disclosure initiatives may complement traditional regulatory approaches under the Kunming-Montreal Framework. Third, the heterogeneity analysis provides a foundation for targeted regulatory strategies to maximize economic efficiency while achieving conservation goals. Future research could extend this work from comparative international perspectives, examine long-term performance implications, investigate additional transmission mechanisms beyond innovation, and assess how biodiversity risk management interacts with broader corporate sustainability strategies such as climate change mitigation and energy transition initiatives. As biodiversity concerns increasingly permeate financial decision-making, understanding these market-based incentive structures are increasingly crucial for corporate strategy and environmental policy design.

#### 6. Research limitations

Despite the valuable insights provided by this study, we acknowledge three noteworthy limitations. First, our NLP-based biodiversity risk management index may be influenced by corporate disclosure tendencies rather than actual risk management practices. Companies could engage in greenwashing, overstating environmental initiatives in annual reports while implementing limited measures in reality. Second, although we employ multiple methods to address endogeneity concerns, we cannot completely rule out potential omitted variables and reverse causality issues. Finally, the study timeframe (2013–2023) encompasses multiple environmental policy shifts that may have imbalanced effects on our results. Future research could adopt more refined risk measurement approaches, extend international comparative perspectives, and explore additional transmission pathways through which biodiversity risk management potentially influences corporate outcomes.

#### CRediT authorship contribution statement

**Wenjie Sun:** Writing – review & editing, Writing – original draft. **Jianhua Mei:** Writing – review & editing, Validation, Formal analysis. **Kecun Chen:** Writing – review & editing, Data curation. **Zhaoqi Li:** Writing – original draft, Writing – review & editing, Methodology, Supervision.

**Table 6**  
Transmission pathway tests.

| Variables      | (1)<br>GI         | (2)<br>TQ          | (3)<br>GI         |
|----------------|-------------------|--------------------|-------------------|
| BD             | 0.085**<br>(2.08) | 0.112*<br>(1.66)   | 0.072*<br>(1.75)  |
| GI             |                   | 0.412***<br>(3.27) |                   |
| IA             |                   |                    | 0.217**<br>(2.44) |
| BD × IA        |                   |                    | 0.088*<br>(1.74)  |
| Controls       | Yes               | Yes                | Yes               |
| Firm FE        | Yes               | Yes                | Yes               |
| Year FE        | Yes               | Yes                | Yes               |
| N              | 10,876            | 10,876             | 10,876            |
| R <sup>2</sup> | 0.325             | 0.396              | 0.341             |

Note: t-statistics are in parentheses, \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.1$ .

## Declaration of competing interest

The authors report there are no competing interests to declare.

## Data availability

The authors do not have permission to share data.

## References

- Adhikari, H.P., More, D.G., Sah, N.B., 2025. Climate change exposure and short-termism: evidence from net trade credit. *Int. Rev. Financ. Anal.* 103, 104193.
- Ambec, S., Cohen, M.A., Elgie, S., Lanoie, P., 2013. The Porter hypothesis at 20: can environmental regulation enhance innovation and competitiveness? *Rev. Environ. Econ. Policy.* 7, 16. [doi:10.1093/reep/res016](https://doi.org/10.1093/reep/res016).
- Bansal, R., Croce, M.M., Liao, W., Rosen, S., 2019. Uncertainty-induced reallocations and growth. *Natl. Bur. Econ. Res. Work. Pap. Ser.*, 26248 <https://doi.org/10.3386/w26248>.
- Cao, Y., Liu, Y., Lu, M., Shan, Y., Zu, Y., 2025. Generosity under environmental pressure: climate change exposure and corporate philanthropy. *Int. Rev. Financ. Anal.* 104, 104336. <https://doi.org/10.1016/j.irfa.2025.104336>.
- Chen, Y., Yin, L., Hu, S., Li, X., 2025. Biodiversity conservation and corporate environmental information disclosure: evidence from a quasi-natural experiment in China. *Int. Rev. Financ. Anal.* 102, 104069.
- Giglio, S., Kuchler, T., Stroebel, J., Zeng, X., 2023. Biodiversity risk. *Natl. Bur. Econ. Res. Work. Pap. Ser.*, 31137 <https://doi.org/10.3386/w31137>.
- Guo, G., Lin, O., Li, Y., Ruan, J., 2024. Corporate carbon emission governance: the mediating role of financial leverage. *Int. Rev. Econ. Finance* 96, 103734.
- He, F., Duan, L., Lucey, B., Hao, J., 2025. Biodiversity risk or climate risk? Which factor affects corporate ESG rating divergence. *Int. Rev. Financ. Anal.* 104, 104302. <https://doi.org/10.1016/j.irfa.2025.104302>.
- Kalhor, M.R., Ahmed, K., 2025. Dynamic linkages and spillover effects of biodiversity risk in socially responsible investment and commodity markets. *J. Environ. Manag.* 374, 124144. <https://doi.org/10.1016/j.jenvman.2025.124144>.
- Kuziboev, B., Ozturk, I., Kalandarov, F., Makhmudov, S., Khodjanizayov, E., 2025. The effect of energy risk on political stability in Europe. *J. Environ. Manag.* 385, 125643. <https://doi.org/10.1016/j.jenvman.2025.125643>.
- Lei, X., Xu, J., Chen, Y., Liu, C., Zhao, K., 2025. Digital Oasis: how green infrastructure is reshaping China's energy resilience landscape. *Systems* 13 (5), 306.
- Lei, X., Xu, X., 2024. Storm clouds over innovation: typhoon shocks and corporate R&D activities. *Econ. Lett.* 244, 112014. <https://doi.org/10.1016/j.econlet.2024.112014>.
- Lei, X., Xu, X., 2025. The "spider web" of Venture Capital: an invisible force driving corporate green technology innovation. *Technol. Soc.* 82 (3), 102882. <https://doi.org/10.1016/j.techsoc.2025.102882>.
- Li, K., Yan, X., Hou, Y., Lv, B., Huang, Y., Liu, J., Li, X., 2024. Can ecological zoning act as an environmental management tool for protecting regional habitat quality: causal evidence from the national key ecological function zone in China. *J. Clean. Prod.* 475, 143623. <https://doi.org/10.1016/j.jclepro.2024.143623>.
- Lin, O., Guan, J., 2024. The impact of media attention, board independence on CEO power, and ESG in state-owned enterprises. *Finance Res. Lett.* 62, 105180. <https://doi.org/10.1016/j.frl.2024.105180>.
- Lin, O., Zhang, X., Guan, J., Lei, X., Deng, L., 2024. Differences in Board Independence's impact on ESG with respect to power constraints: evidence from a heterogeneity perspective. *Pol. J. Environ. Stud.* 33 (4), 3771–3782. <https://doi.org/10.15244/pjoes/177165>.
- Liu, W., Liu, Y., Wen, F., Gong, X., 2025. Climate policy uncertainty and analyst earnings forecasts: evidence from the chinese energy sector. *Int. Rev. Financ. Anal.* 100, 103960.
- Ma, F., Wu, H., Zeng, Q., 2024. Biodiversity and stock returns. *Int. Rev. Financ. Anal.* 95, 103386. <https://doi.org/10.1016/j.irfa.2024.103386>.
- Marrero, G.A., Puch, L.A., Ramos-Real, F.J., 2015. Mean-variance portfolio methods for energy policy risk management. *Int. Rev. Econ. Finance* 40, 246–264. <https://doi.org/10.1016/j.iref.2015.02.013>.
- Ocasio, W., 1997. Towards an attention-based view of the firm. *Strateg. Manag. J.* 18 (S1), 187–206.
- Ongsakul, V., Chatjuthamard, P., Chintrakarn, P., Jiraporn, P., 2025. Climate change exposure in uncertain times: a text-based approach. *Int. Rev. Econ. Finance* 100, 104083. <https://doi.org/10.1016/j.iref.2025.104083>.
- Tan, R., Hou, K., Wu, H., 2025. The cost of biodiversity protection: national key ecological functional zone and labor demand in China. *J. Asian Econ.* 97, 101895. <https://doi.org/10.1016/j.asieco.2025.101895>.
- Teece, D.J., 2018. Dynamic capabilities as (workable) management systems theory. *J. Manag. Organ.* 24 (3), 359–368.
- Wang, Y., 2025. The silent cost of biodiversity loss: unveiling its impact on institutional ownership. *Int. Rev. Financ. Anal.* 103, 104075. <https://doi.org/10.1016/j.irfa.2025.104075>.
- Yang, C., Song, D., 2025. How does climate change exposure affects green innovation in China: a framework for categorizing innovation. *Int. Rev. Financ. Anal.* 103, 104253. <https://doi.org/10.1016/j.irfa.2025.104253>.

- Zhang, L., Luo, Z., Yu, X., Yang, Q., Wang, J., 2025. Economic policy uncertainty and corporate green innovation. *Int. Rev. Financ. Anal.* 97, 103797.
- Zhang, N., Han, H., 2025. The new environmental protection law, political connections and corporate ESG performance. *Int. Rev. Financ. Anal.* 102, 104110.
- Zhao, Y., Qu, X., Lucey, B., 2025. Managerial myopia and biodiversity alignment- evidence from China. *Int. Rev. Financ. Anal.* 100, 103974. <https://doi.org/10.1016/j.irfa.2025.103974>.
- Zhou, Y., Lucey, B.M., He, F., 2025. Dividend payouts and biodiversity risk - Chinese evidence. *Res. Int. Bus. Finance.* 76, 102792. <https://doi.org/10.1016/j.ribaf.2025.102792>.